

Progress Report: 7/22/26

Satellite Green Vegetation Fraction (GVF) vs ground PhenoCam at GBOV/NEON site

Project goal (from internship description)

GVF is the fraction of horizontal land surface covered by photosynthesizing vegetation. Prior work found that NOAA satellite/model GVF greenup is slower and more gradual than in-situ data, while ground transitions look more abrupt. This step compares another satellite GVF product (GBOV provided timeseries) to PhenoCam GCC and NDVI at the same NEON/GBOV sites, and extracts SOS/MOS/DOS/EOS dates plus agreement scores.

What I did this week

- Loaded daily satellite GVF files from validation_pipeline/input/GBOV_2023 and GBOV_2024 (format YYYY/MM/DD, GVF, values integers 0-100).
- Matched each bare NEON site id to its PhenoCam ROI and fetched GCC_90 + NDVI_90.
- Built 3 curve plots (GVF + GCC + NDVI) with SOS/MOS/DOS/EOS markers for all three series (shared/plot_satellite.py).
- Computed new agreement scores: GVF vs GCC and GVF vs NDVI (phenophase gap days + DTW/step → divergence; lower = better).
- Result: plotted 23 of 24 site-years (STER 2024 skipped - no PhenoCam data that year). Plots in validation_pipeline/output/GBOV_2023/ and GBOV_2024/.

What the overlays show

- At many sites (esp. deciduous forest like BART), PhenoCam GCC/NDVI rise and fall more abruptly; satellite GVF is smoother / more rounded. That matches the internship note that in-situ transitions look sharper than satellite/model GVF.
- Mean SOS DOY is later for GVF (68) than GCC (55) or NDVI (52). Satellite greenup often starts after the camera.
- **(median):** GVF vs GCC gap 32 d, divergence 2.34; GVF vs NDVI gap 32 d, divergence 2.32. Typical offset is about 3-5 weeks across the four phases.
- Deciduous forest (DB) agrees best (mean GVF-NDVI gap ~21 d). Grassland (GR) and agriculture (AG) disagree more (gaps ~58 / 68 d). Shrub (SH) is also relatively good at SRER/ONAQ.
- lowest GVF-NDVI divergence: SRER 2024, BART 2024, HARV 2024, HARV 2023, BART 2023.
- Hardest site-years: STER 2023 (AG, GCC phases failed), CPER 2024 and JORN 2023/2024 (grasslands with large phase offsets).

Overview

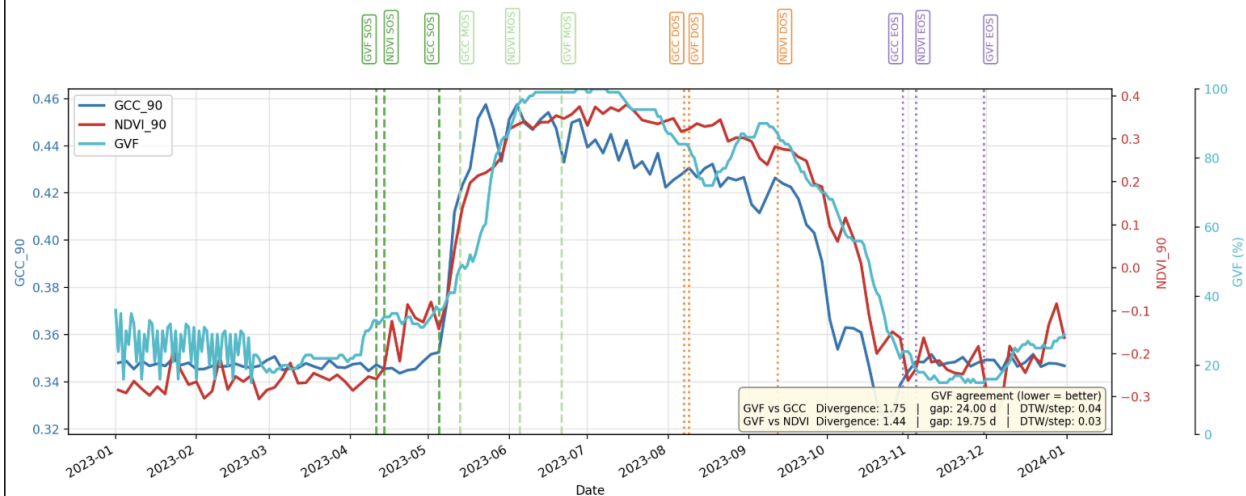
These GBOV GVF overlays are a first higher resolution / alternate satellite check against ground phenology (PhenoCam), not yet against NOAA 1 km/4 km GVF or the weather model GVF.

Early takeaway: the gradual satellite greenup vs abrupt ground transitions shows up here too, and agreement is strongest at deciduous NEON sites. Next steps would be

- (1) bring in NOAA GVF at the same points for a three-way compare, and
- (2) if time allows, derive cover from higher res imagery for further validation.

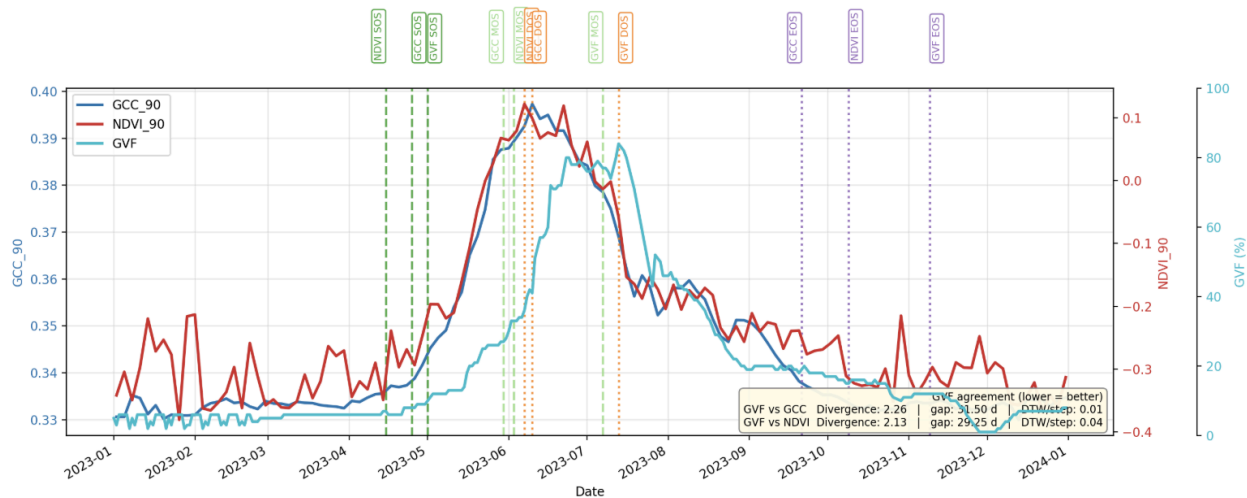
Example overlays

Satellite GVF vs PhenoCam GCC_90 & NDVI_90 (NEON.D01.BART.DP1.00033_DB_1000, 2023)



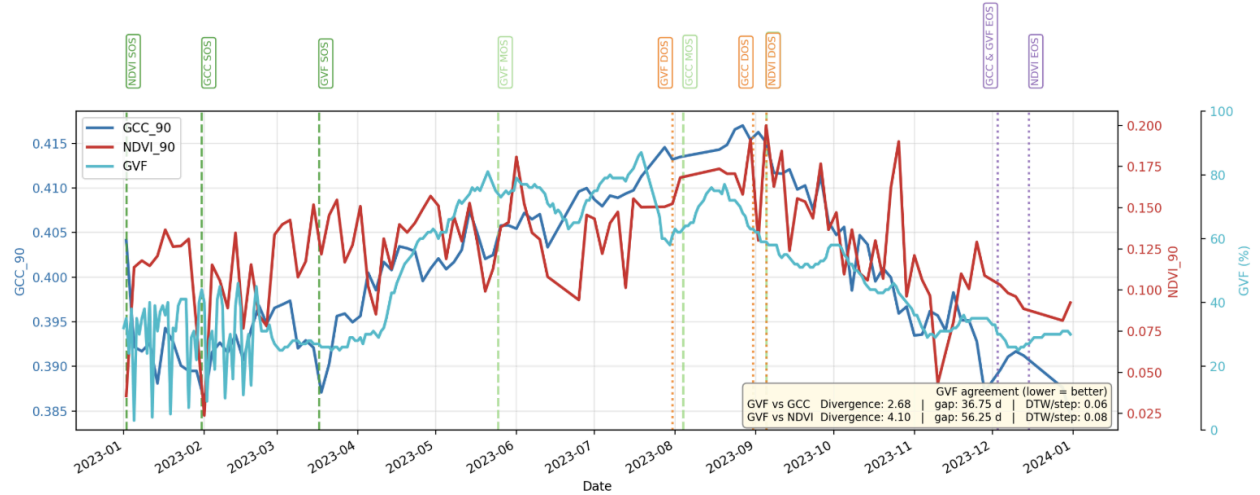
BART 2023 - deciduous forest, stronger GVF-ground agreement

Satellite GVF vs PhenoCam GCC_90 & NDVI_90 (NEON.D10.CPER.DP1.00033_GR_1000, 2023)



CPER 2023 - grassland, GVF lags PhenoCam greenup

Satellite GVF vs PhenoCam GCC_90 & NDVI_90 (NEON.D08.TALL.DP1.00033_EN_1000, 2023)



TALL 2024 - evergreen, smoother satellite GVF vs abrupt camera

All site-years ranked by GVF vs NDVI divergence

Lower divergence = better agreement. Gap = mean |DOY| difference across SOS/MOS/DOS/EOS. STER 2024 omitted (no PhenoCam).

#	Site	Yr	Veg	Gap GCC	Div GCC	Gap NDVI	Div NDVI
1	SRER	2024	SH	8.5	0.67	9.8	0.74
2	BART	2024	DB	19.8	1.44	10.8	0.81
3	HARV	2024	DB	16.5	1.22	11.0	0.82
4	HARV	2023	DB	74.8	5.39	19.2	1.44
5	BART	2023	DB	24.0	1.75	19.8	1.44
6	SRER	2023	SH	8.0	0.65	21.5	1.62
7	DELA	2023	DB	35.8	2.58	23.5	1.70
8	ORNL	2024	DB	26.8	1.97	23.2	1.71
9	DELA	2024	DB	16.5	1.21	25.2	1.84
10	CPER	2023	GR	31.5	2.26	29.2	2.13
11	KONA	2023	AG	26.0	1.88	30.2	2.19
12	MOAB	2024	GR	32.5	2.42	31.5	2.32
13	ONAQ	2024	SH	24.0	1.75	34.8	2.56
14	ORNL	2023	DB	40.0	2.89	32.5	2.59
15	ONAQ	2023	SH	27.0	1.99	36.5	2.70

16	MOAB	2023	GR	42.8	3.13	47.8	3.48
17	KONA	2024	AG	47.8	3.48	52.8	3.81
18	TALL	2024	EN	71.0	5.11	54.2	3.94
19	TALL	2023	EN	36.8	2.68	56.2	4.10
20	JORN	2024	GR	47.2	3.55	70.2	5.15
21	JORN	2023	GR	69.0	5.13	82.5	6.02
22	CPER	2024	GR	90.8	6.52	89.5	6.47
23	STER	2023	AG	n/a	n/a	122.5	8.89

Veg codes: DB = deciduous broadleaf, EN = evergreen needle, GR = grassland, AG = agriculture, SH = shrub.